

Perfect security—an introduction to information-theoretic security

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(introduced by C. Shannon in 1949)

- Goals:
- Formalise what "perfect security" of a cryptosystem is.
 - Characterise perfectly secure cryptosystems.
 - Construct examples of perfectly secure cryptosystems
... only to find them of limited practical value.
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Assumptions:

- We consider a fixed cryptosystem with sets P, C, K of plaintexts, ciphertexts, and keys, resp., and encryption and decryption functions e_k and d_k as before.
- P, C , and K are all non-empty and finite.
- We randomly and independently choose plaintexts $x \in P$ and keys $k \in K$ according to some probability distributions on P and K .

Having picked x and k , we encrypt $y = e_k(x)$.

We write X, Y, K for random variables corresponding to plaintexts, ciphertexts, and keys.

(Recall that informally, a random variable is a quantity that depends on the outcome of a random process. Here: choose (x, k) .)

- We assume that $\text{Prob}(X=x), \text{Prob}(Y=y) > 0$ for all $x \in \mathcal{P}$ and $y \in \mathcal{C}$.

Q: How can we express $\text{Prob}(Y=y)$ in terms of the given probabilities on $\mathcal{P}$ and $\mathcal{K}$?

Exercise: Write $e_k[\mathcal{P}] := \{e_k(x) : x \in \mathcal{P}\}$.

Then for each $y \in \mathcal{C}$, we have

$$\text{Prob}(Y=y) = \sum_{\substack{k \in \mathcal{K} \\ \text{s.t. } y \in e_k[\mathcal{P}]}} \text{Prob}(K=k) \cdot \text{Prob}(X=d_k(y)).$$

Reminder: For events A, B with $\text{Prob}(B) > 0$, the conditional probability of A given B is

$$\text{Prob}(A|B) := \frac{\text{Prob}(A \cap B)}{\text{Prob}(B)}$$

In other words, $\text{Prob}(A|B)$ is the probability of A occurring assuming that B occurred.

Defn Our cryptosystem (satisfying the assumptions) from above is perfectly secure if

$$\text{Prob}(X=x | Y=y) = \text{Prob}(X=x)$$

for all $x \in \mathcal{P}$ and $y \in \mathcal{C}$.

In other words, perfect security means that knowing the ciphertext reveals nothing about the plaintext.

Q: Are there any perfectly secure cryptosystems?

Lemma Our cryptosystem is perfectly secure iff

$$\text{Prob}(Y=y \mid X=x) = \text{Prob}(Y=y)$$

for all $x \in \mathcal{P}$ and $y \in \mathcal{C}$. (Note the difference!)

Proof: In general, if $\text{Prob}(A), \text{Prob}(B) > 0$, then

$$\text{Prob}(A \mid B) \cdot \text{Prob}(B) = \text{Prob}(A \cap B) = \text{Prob}(B \mid A) \cdot \text{Prob}(A)$$

Hence:

$$\text{Prob}(Y=y \mid X=x) \stackrel{!}{=} \frac{\text{Prob}(X=x \mid Y=y)}{\text{Prob}(X=x)} \cdot \text{Prob}(Y=y)$$

$$= \text{Prob}(Y=y)$$

$$\Leftrightarrow \text{Prob}(X=x \mid Y=y) = \text{Prob}(X=x) \parallel$$

Propn Suppose that our cryptosystem is perfectly secure.

Then:

$$|\mathcal{K}| \geq |\mathcal{C}| \geq |\mathcal{P}|$$

(That is, #keys $\geq$ #ciphertexts $\geq$ #plaintexts.)

Proof:

- Let S and T be finite sets.

TFAE: — $|S| \leq |T|$

- There exists an injective function $S \rightarrow T$.
- There exists a surjective function $T \rightarrow S$.

Pf: exercise!

- Pick any $k \in \mathcal{K}$. (We can do this since we assumed $\mathcal{K} \neq \emptyset$!)

Since $d_k(e_k(x)) = x$ for all $x \in \mathcal{P}$, e_k is an injective function $\mathcal{P} \rightarrow \mathcal{C}$. Indeed, if $e_k(x) = e_k(x')$, then $x = d_k(e_k(x)) = d_k(e_k(x')) = x'$.

Hence, $|\mathcal{P}| \leq |\mathcal{C}|$. (Note that we didn't use perfect security yet!)

- Let $x \in \mathcal{P}$ and $y \in \mathcal{C}$.

Then: $\text{Prob}(Y=y | X=x) = \text{Prob}(Y=y) > 0$

perfect security
+
lemma

by assumption

In particular: $\forall x \in \mathcal{P} \forall y \in \mathcal{C} \exists k \in \mathcal{K}: y = e_k(x)$

Hence, for any fixed choice of $x \in \mathcal{P}$ ($\neq \emptyset!$), the function

$$\mathcal{K} \longrightarrow \mathcal{C}, k \mapsto e_k(x)$$

is surjective. $\leadsto |\mathcal{K}| \geq |\mathcal{C}|$ //

Our main result on perfectly secure cryptosystems is the following:

Shannon's Theorem: Suppose that $|\mathcal{P}| = |\mathcal{C}| = |\mathcal{K}|$,

in addition to our assumptions from before.

Then our cryptosystem is perfectly secure if and only if

$$(a) \text{ Prob}(K = k) = \frac{1}{|\mathcal{K}|} \text{ for all } k \in \mathcal{K}$$

(i.e. Keys are chosen uniformly at random)

and

$$(b) \forall x \in \mathcal{P} \forall y \in \mathcal{C} \exists! k \in \mathcal{K}. e_k(x) = y$$

Proof:

"only if": Suppose that our cryptosystem is perfectly secure.

We know from the proof of the preceding proposition that for each $x \in \mathcal{P}$, the function

$$\mathcal{K} \rightarrow \mathcal{C}, k \mapsto e_k(x)$$

is surjective. Since $|\mathcal{K}| = |\mathcal{C}| < \infty$, it is also injective ($\rightarrow$ Pigeonhole principle), hence bijective.

This proves (b)!

For (a), fix $y \in \mathcal{C}$. Enumerate $\mathcal{P} = \{x_1, \dots, x_n\}$ and $\mathcal{K} = \{k_1, \dots, k_n\}$, where $n = |\mathcal{K}| = |\mathcal{P}| = |\mathcal{C}|$

and $\boxed{e_{k_i}(x_i) = y}$.

This is possible since $\mathcal{K} \rightarrow \mathcal{P}, k \mapsto d_k(y)$ is bijective. (Indeed, for $x \in \mathcal{P}$ and $y \in \mathcal{C}$, we already know that there exists $k \in \mathcal{K}$ with $e_k(x) = y$ and thus $x = d_k(e_k(x)) = d_k(y)$. That is, $\mathcal{K} \rightarrow \mathcal{P}, k \mapsto d_k(y)$ is surjective hence bijective.)

Since $|P| = |C|$, $d_k = e_k^{-1}$ and $e_k = d_k^{-1}$ for each $k \in \mathcal{K}$ and after enumerating $\mathcal{K} = \{k_1, \dots, k_n\}$, we may take $x_i = d_{k_i}(y)$ above.

Now:

$$\text{Prob}(X = x_i) \stackrel{?}{=} \text{Prob}(X = x_i | Y = y)$$

↑
perfect security

$$\stackrel{?}{=} \frac{\text{Prob}(Y = y | X = x_i)}{\text{Prob}(Y = y)} \text{Prob}(X = x_i)$$

↑
see above

$$\stackrel{?}{=} \frac{\text{Prob}(K = k_i) \cdot \text{Prob}(X = x_i)}{\text{Prob}(Y = y)}$$

$$\leadsto \text{Prob}(K = k_i) = \text{Prob}(Y = y) > 0 \text{ for all } i \text{ !}$$

$$\leadsto \text{Prob}(K = k_i) = \frac{1}{n}.$$

"if": omitted, see e.g. Smart, Thm 5.4. //